ChatGPT

Executive Summary

We propose a **unified prompt-generator** system: a tool where users (prompt engineers, indie devs, game designers, product managers, educators, etc.) can mix-and-match proven prompt formulas and templates via an intuitive interface. Users pick their persona/role, desired output type (app, tool, game, script, marketing copy, etc.), and writing "ingredients" (tone, constraints, examples, etc.). The system stitches these into a high-quality, domain-tailored prompt and also translates it into **Just Plain English (JPE)** so anyone can understand the request. This report surveys top prompt frameworks (AIDA, SCQA, PASTOR, role+constraints+examples, Chain-of-Thought, Tree-of-Thought, etc.), compares UI patterns (wizard, templates, advanced toggles), and defines metrics (quality, creativity, specificity, reproducibility, safety) with an evaluation plan. We provide example schemas, Lua code stubs, interface flowcharts, and a prioritized development roadmap (MVP to full product). The goal is an "elite prompt lab" that's both powerful and user-friendly, letting creators press a button and get the *best* possible AI prompt for their task.

Target Users and Personas

Our tool serves a broad range of knowledge workers and creatives:

- **Prompt Engineers / AI Researchers:** Need to quickly construct precise prompts for testing LLMs or prototypes. They'll appreciate full access to advanced parameters, custom schemas, and plugin/extensions support.
- **Indie Developers & Game Designers:** Want to generate code snippets, game story ideas, or mechanics descriptions. They look for simple templates ("game concept", "level design prompt") and perhaps Lua code output (since Lua is popular in game scripting like LÖVE or Roblox).
- **Product Managers / Marketers:** Need marketing copy, feature ideas, or user flows. They use formulas like AIDA/PASTOR and want straightforward outputs (e.g. bullet lists, JSON).
- **Educators / Students:** Teach AI or creative thinking. They need a gentler interface, examples, and the JPE explanation to demystify AI prompts.
- **General Power Users:** Tech-savvy folks who want creative bots (story writers, hobbyists) might use "chat-like" interfaces or templates.

Each persona can customize prompts, but we assume everyone benefits from: role/context settings, output formats, examples, and style constraints. For instance, a game designer (persona) might pick a "Game Design Template", set role = "indie dev game designer", output = "game concept JSON", and get a prompt ready for GPT-4 or another model.

Supported Output Types

The system's output is flexible. By default, it generates **text prompts** (for AI models). But power users often need structured or code outputs:

- **Text prompts (plain or JPE)** for any creative or research task.

- **Structured JSON** (e.g. `{"task":"make game","genre":"platformer","features":[...]}`) to feed into pipelines or other tools.
- **Lua code snippets** or other language templates. Since the user likes Lua, we include Lua snippet generation (for scripts, game logic, plugin stubs, etc.).
- **Scripts in other languages** (e.g. Python, JavaScript) or format templates (Markdown, HTML) may be supported via plugins.
- **Export options:** Copy-to-clipboard for the prompt/JPE, or export as .txt/.json/.lua files.

Thus the UI will let users choose an output format (plain text, JSON, Lua, etc.) per prompt.

Survey of Prompt-Engineering Frameworks

We reviewed popular prompt formulas and structures used across industries. Below is a comparative table of key formulas, with use-cases, pros, and cons (each formula also yields different prompt patterns):

Formula/Pattern	Description & Use-Cases	Pros	Cons
AIDA (Attention–Interest–Desire–Action) [1]	A classic copywriting sequence: grab *Attention*, spark *Interest*, build *Desire*, end with *Action*. Used for landing pages, ads, emails [2].	Time-tested for persuasive copy; gives clear sequence [1]; works in many contexts.	Can feel formulaic; may overlook depth of *Desire* [3]; relies on a strong hook.
PASTOR (Problem–Amplify–Story/Solution–Transformation–Offer–Response) [4]	A narrative sales framework: identify the Problem, amplify pain, share Solution story, show Transformation, present Offer, then Response (CTA). Good for long-form marketing (sales pages, launches) [4].	Builds trust via storytelling and social proof [5]; thorough persuasion structure.	Lengthy; can overwhelm with info [6]; requires testimonials or detailed examples; more than needed for simple prompts.
SCQA (Situation–Complication–Question–Answer) [7]	Business storytelling: set the Situation (context), introduce a Complication (problem), pose a central Question, provide an Answer. Useful for analysis tasks, pitches, or explainer prompts.	Structures logical flow clearly [7]; ensures context before request.	May feel formal; overkill for casual/chat tasks; sometimes rigid narrative tone.
Role + Constraints + Examples	A generic meta-format: specify *Role* ("You are a Python expert"), list *Constraints* (tone, length, output format), and provide *Examples* of desired output. Common best-practice template [8].	Very flexible; guides model explicitly; examples anchor expectation [8].	Prompts can become long/verbose; writing good examples is manual work; model may ignore constraints unless clearly delimited.

Formula/Pattern	Description & Use-Cases	Pros	Cons
Chain-of-Thought (CoT) [9]	Instruct model to "think step by step" by explicitly including reasoning steps. Originally for math/logic tasks, but generalizable (e.g. "You are a detective, explain your reasoning").	Improves complex reasoning and accuracy [10] [9]; useful when intermediate steps matter.	Increases token usage (longer prompts & outputs); not needed for simple tasks; some models are inconsistent even with CoT.
Tree-of-Thought (ToT) [11]	Extends CoT by branching multiple solution paths (like exploring alternatives). More agentic: the model may "backtrack" and try different options.	Allows exploring creative or complex solutions; mimics human trial/error [11].	Even more complex and resource-intensive; requires custom prompting or loop logic; not native to simple chat interfaces.
Other Copy Frameworks (PAS, QUEST, BAB, etc.)	e.g. PAS (Problem–Agitate–Solution), BAB (Before–After–Bridge), QUEST. These are often subsets or variations of above patterns.	Well-known in marketing/writing; good for specific formats (email, ads, tutorials).	Similar pros/cons as AIDA/PASTOR; often narrow use-case; risk of sounding like formula.

Notes: All these can be embedded into our tool as *prompt templates*. For example, the AIDA formula can have a UI section for each letter (Attention:, Interest:, etc.), or fillable slots in a macro-prompt. We'd cite best practices: e.g. AIDA has 120+ years of proven results [2] and is easy to implement [12]; SCQA ensures logical flow [7]; role/examples format is widely recommended [8]; CoT/ToT boost reasoning [9] [11]. Each formula's pros/cons help a user pick the right one (the UI might suggest "Use AIDA for marketing copy, CoT for solving puzzles," etc.).

UI/UX Design Patterns

Designing the prompt-builder UI is key for usability. Common patterns include:

- **Wizard/Stepper:** Multi-step form guiding users through sections (select persona → choose template → enter details → generate). *Pros:* Simplifies decision at each step; novice-friendly. *Cons:* Many clicks; hard to see context of all choices at once.
- **Single-page Form with Sections:** All options visible in sections (like a web form with headings: *Role*, *Goal*, *Tone*, *Constraints*, etc.). *Pros:* Fast editing, instant preview of combined prompt; can handle advanced settings. *Cons:* Overwhelming for new users; too many fields on screen.
- **Template/Accordion Panels:** Provide templates or presets (e.g. "E-commerce Ad," "Game Concept") in a sidebar or tabs. Selecting a template auto-fills sections. *Pros:* Jump-starts prompts for common tasks; copy-to-clipboard easily accessible. *Cons:* Limits flexibility if user's task is novel; may clutter UI with many options.

- **Progressive Disclosure:** Hide advanced or rare options under an “Advanced” toggle, with minimal initial view [13]. For instance, basic fields (task, tone) show first; expert fields (temperature, stop tokens) appear after clicking “Show advanced.” *Pros:* Keeps UI simple for beginners; yet powerful features available. *Cons:* Users might miss hidden options; context can be lost when revealing them.
- **Chat-like Interface with Suggestions:** User types a rough prompt, and the tool suggests improvements or expansions (like copy editing). *Pros:* Very natural for users familiar with chatbots; supports iterative refinement. *Cons:* Harder to enforce structure; less transparent for novices.
- **Copy-to-Clipboard and Export Buttons:** Always provide buttons to copy the generated prompt or JPE, and export JSON/Lua. This is an action pattern (not layout) but critical for usability.
- **Real-time Preview/Result Pane:** Show the generated prompt (and its JPE translation) side-by-side with the input controls. *Pros:* Immediate feedback; user learns what selections do. *Cons:* Requires dynamic updating, may be complex to implement.

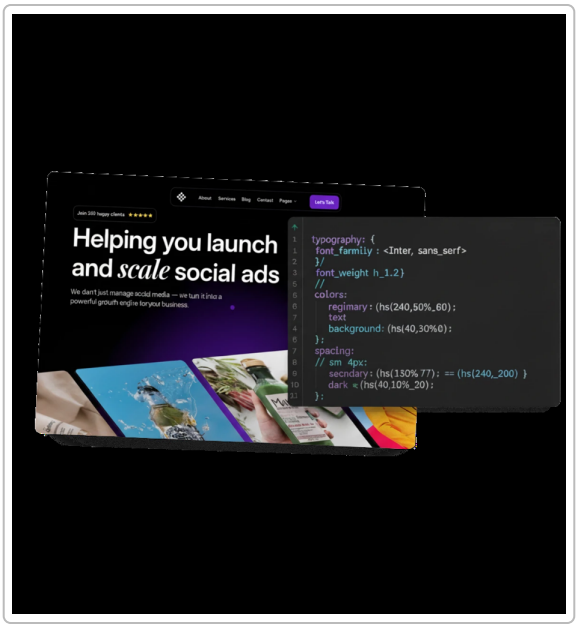


Figure: Example “Prompt Builder” UI (screenshot of a design tool with controls on one side and generated code/outputs on the other). A split-view layout (input on left, output on right) is a common pattern. This example shows how formatting (color palettes on left, code snippet on right) can reflect a template-driven workflow (here, generating a design system from an image prompt). In our tool, we might similarly display the assembled prompt or JSON next to the user’s settings panel.

Overall, an **intuitive interface** might combine a left-side navigation (personas/templates) and right-side detail forms, with a fixed “Generate” button. We’d follow UX best practices: use bullet points or numbered steps for clarity [14], group related fields visually, label everything clearly (e.g. “Tone: Professional vs Casual”), and ensure tooltips or examples appear for ambiguous terms. Progressive disclosure [13] is crucial: hide parameters (like model temperature, or “analysis mode” vs “concise mode”) behind toggles. Copy-to-clipboard icons should be prominent near any output textbox (the NNGroup even advises adding obvious copy icons for ease [13]). Templates or presets could be presented as clickable cards or a dropdown menu.

UI Patterns Table (Examples):

Pattern	Description	Pros	Cons
Wizard (Multi-step)	Linear steps (select persona → template → details → review).	Low cognitive load per step; guided process.	Cumbersome for experts; hard to jump around; many clicks.
One-Page Form	All fields on one scrolling page (with sections).	Fast for experienced users; see all options at once.	Intimidating for newbies; risk of messy layout.
Accordion/ Tabs	Collapsible sections or tabbed groups of fields.	Organizes related inputs; can collapse advanced parts.	Hiding content may confuse users; discoverability issues.
Template Gallery	Pre-made prompt templates/presets to choose from.	Quick start; shows examples; user learns by example.	Templates may not cover edge-cases; encourages copying.
Chat-style UI	Conversational input with AI refining the prompt.	Very natural; iterative refinement possible.	Hard to enforce strict constraints; might undermine structure.
Copy/Export Actions	Buttons/icons for "Copy Prompt", "Copy JPE", "Export JSON/Lua".	Makes results easily transferable; meets dev needs.	(No downside—these are essential tools.)
Preview Pane	Live-updating pane showing the composed prompt/JPE.	Immediate feedback; catches errors early.	Requires real-time logic; could confuse with partial output.

We will incorporate patterns like modals or tooltips for examples and constraints. The progressive disclosure pattern is especially advised by NNGroup: e.g., have an "Advanced options" button revealing less-used fields [13]. Likewise, use formatting (bullets, bold, code blocks) inside the prompt builder to highlight fixed templates vs user-entered text [14]. For instance, prompts can be built up with markdown bullets or sections, making the generation more robust.

Unified Prompt Schema and Templates

From the above survey, we design a **unified prompt schema** that accommodates all key parts. We recommend a JSON-like internal representation with fields such as:

- `persona` / `role`: The user's or model's role (e.g. "You are a friendly marketing guru").
- `task`: The core objective or task instruction (e.g. "Generate 5 email subject lines...").
- `context` / `background`: Any background info or constraints (domain, audience, goals).
- `tone` / `style`: Desired tone, formality, and style guidelines (e.g. "concise, enthusiastic, grade-8 reading level").
- `constraints`: A list of rules (e.g. "max 100 words", "no jargon", "use JSON output").
- `format`: The expected output format (e.g. "Return result in Markdown table").
- `examples`: Few-shot examples as text pairs, if using example-based prompting.

- `formula`: Chosen framework (AIDA, SCQA, etc.) or structure to apply.
- `extension`: Extra modules/plugins (e.g. translation, summarization).
- **Output fields** like `prompt_text` and `JPE_text` (generated by the system from the inputs).

An example unified schema in JSON form might look like:

```
{
  "persona":   "game developer",
  "role":      "indie studio creative director",
  "task":      "Design a new mobile endless runner game concept",
  "context":   "Target audience: casual gamers, ages 12-18.",
  "tone":      "upbeat and clear",
  "constraints": ["Include 3 unique game mechanics", "Answer in bullet points",
"Max 200 words"],
  "format":    "JSON {\"concept\":..., \"mechanics\":..., \"art_style\":...}",
  "examples": [
     { "input": "Puzzle game idea", "output": "A grid-based puzzle game
where..." }
  ],
  "formula":   "role+examples",
  "additional": { "engine": "gpt-4o", "temperature": 0.7 }
}
```

From this schema, the tool programmatically assembles the actual prompt. **Example prompt template for a game concept**:

> *Prompt (structured)*:
> *Persona:* "You are a creative game designer."
> *Task:* "Design an endless runner game for mobile devices."
> *Context:* "It should appeal to kids and use cartoon graphics."
> *Output Format:* "Provide JSON with keys `concept`, `mechanics`, `theme`."
> *Constraints:* "Max 150 words. Mention 3 unique mechanics."
> *Example:* "(Optional: include an example of a similar game idea if helpful.)"
>
> *Generated Prompt:*
> "You are a creative game designer. Design an endless runner game for mobile devices, appealing to kids with cartoon graphics. Provide the answer as JSON with keys `concept`, `mechanics`, and `theme`. Mention 3 unique gameplay mechanics, and keep the description under 150 words."

Then the system would also produce **JPE** (plain language) for this prompt:

> *JPE Explanation:*
> "Ask the AI to pretend it's a game designer and come up with a fun endless runner game idea for mobile. Make it kid-friendly with cartoon style. Say what the game is about (`concept`),

describe three game mechanics (`mechanics`), and the overall theme. Write the answer in JSON format and keep it short (under 150 words).”

We’d present recommended templates for various domains. For example, **App generation template** (targeting productivity tool) might specify `role: "productivity app expert"` , `tone: "helpful"` , and ask for features list or UI sketches. **Tool/script template** (e.g. “generate a Lua plugin script for …) would use a `role` like “Lua programmer” and `format: "Lua code"` . Each template pre-fills parts of the schema, with placeholders for user details.

Examples (short illustrations):

- **App Example (To-Do List Generator):**

Schema:

• persona/role: “productivity coach”
• task: “Create a to-do list app feature list.”
• audience: “busy professionals.”
• output: “Return JSON array of feature descriptions.”
• constraints: “Each feature ≤2 sentences.”

Generated Prompt:

> “You are a productivity coach. List 5 features for a to-do list app targeting busy professionals. Provide the answer as a JSON array of objects with `feature` and `benefit` fields. Each entry should be concise (max 2 sentences).”

JPE Explanation:

> “Ask the AI (acting as a productivity coach) to suggest five features for a new to-do list app for busy professionals. Output those features as a JSON list, where each item has a short name and a sentence about why it's useful. Each sentence should be no more than two lines.”

- **Tool Example (Lua Code Snippet):**

Schema:

• persona/role: “Lua scripting expert”
• task: “Generate a Lua function to reverse a string.”
• format: “Lua code block.”
• constraints: “Include comments. Use standard Lua, no external libraries.”

Generated Prompt:

> “You are an expert in Lua programming. Write a Lua function called `reverseString` that takes a string and returns it reversed. Include comments explaining each step. Use only core Lua (no libraries).”

JPE Explanation:

> "Tell the AI (as a Lua expert) to write a function `reverseString` in Lua. The function should take an input string and return it backwards. We want comments in the code explaining what each part does, and use only Lua's built-in features."

- **Game Example (Endless Runner):** (see schema above).

These templates ensure consistency and save time. The actual system would display example filled JSON or prompt text in the UI after the user inputs specifics.

Lua Script Examples

Below are illustrative Lua snippets for generating prompts and producing JPE output.

1. Prompt Generation in Lua: Suppose we build a simple function that composes a prompt string from user inputs (persona, task, etc.). In a real app, these values come from UI fields.

```lua
-- Compose a prompt string based on fields
function compose_prompt(persona, task, context, constraints, format)
  local prompt = ""
  if persona then prompt = prompt .. "You are a " .. persona .. ". " end
  if task then prompt = prompt .. task end
  if context then prompt = prompt .. " " .. context end
  if format then prompt = prompt .. " Provide the answer in " .. format .. "."
end
  if constraints and #constraints > 0 then
    prompt = prompt .. " Constraints: " .. table.concat(constraints, "; ") ..
"."
  end
  return prompt
end

-- Example usage
local persona = "game designer"
local task = "Generate an endless runner game concept for mobile devices."
local context = "It should appeal to children and use colorful cartoon
graphics."
local constraints = {"Include 3 unique mechanics", "Answer in bullet points",
"Max 150 words"}
local format = "JSON with keys {concept, mechanics, art_style}"
local prompt_text = compose_prompt(persona, task, context, constraints, format)
print(prompt_text)
```

This would output something like:

```
You are a game designer. Generate an endless runner game concept for mobile devices. It should appeal to children and use colorful cartoon graphics. Provide the answer in JSON with keys {concept, mechanics, art_style}. Constraints: Include 3 unique mechanics; Answer in bullet points; Max 150 words.
```

2. Generating JPE (Just Plain English): In practice, this might call an AI model to paraphrase the prompt. Here's a stub:

```
-- Mock function to get an explanation via an AI model
function explain_prompt_in_jpe(prompt_text)
  local instruction = "Explain the following prompt in simple English, step by step:"
  local ai_input = instruction .. "\n" .. prompt_text
  -- Here we'd call an AI API with ai_input. We'll stub it:
  return "Ask the assistant to act as a game designer and describe a new endless runner game for kids. Make it colorful and mention three cool mechanics. Output the plan in JSON format with fields for the concept, mechanics, and art style."
end

-- Example
local jpe = explain_prompt_in_jpe(prompt_text)
print(jpe)
```

In a real implementation, `explain_prompt_in_jpe` would call the chosen LLM (via OpenAI, Azure, or a local model) to paraphrase the prompt. The result is a readable description of the prompt's intent. For example, the above might yield:

> "Ask the AI to pretend it's a game designer. Have it describe a fun endless runner game for kids with cartoon graphics, and list three unique gameplay mechanics. Tell it to output the ideas in JSON with keys `concept`, `mechanics`, and `art_style`."

These snippets would be part of the backend logic. The actual code might use an HTTP client or OpenAI Lua SDK to make requests. The UI (e.g. in LÖVE or a web frontend) would capture user inputs, run similar functions, and display the results.

UI Wireframe and Interaction Flows

The user's journey through the interface can be visualized as follows:

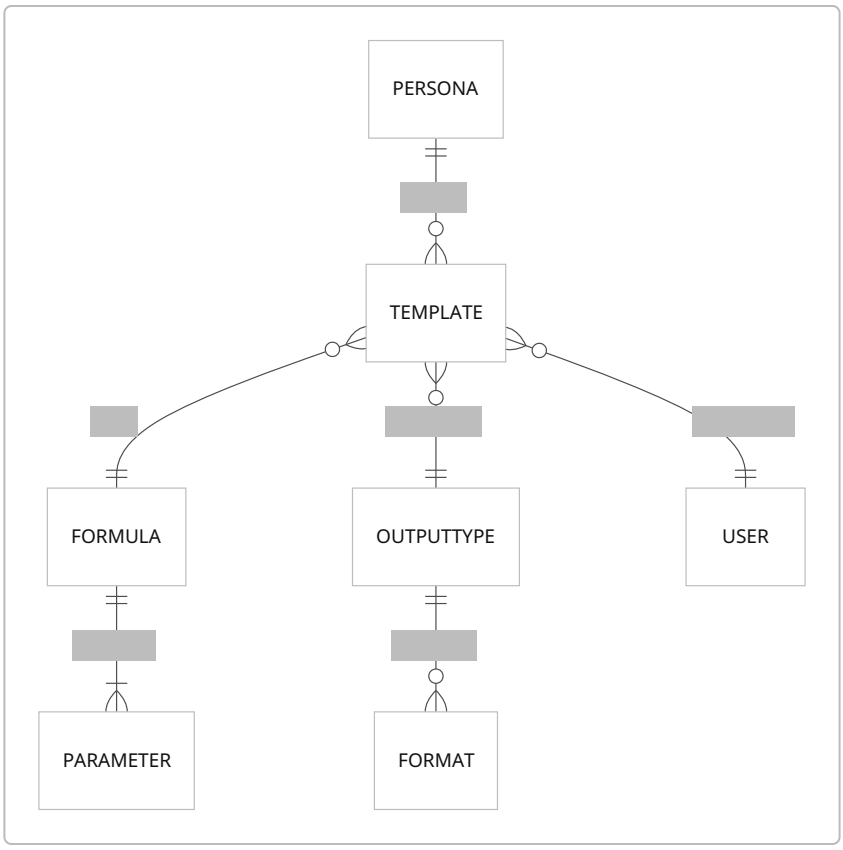
```
flowchart TD
    A[Start: Select Persona/Template] --> B{Choose Output Type};
    B --> C[Configure Details (Role, Tone, Context)];
```

```
    C --> D[Set Constraints & Examples];
    D --> E[Press "Generate Prompt"];
    E --> F[Display Generated Prompt];
    F --> G[Press "Explain in JPE"];
    G --> H[Display JPE Output];
    F --> I[Copy/Export Options];
    I --> J[Done];

    style A fill:#E8F0FE,stroke:#000,stroke-width:1px
    style F fill:#E8F0FE,stroke:#000,stroke-width:1px
```

- **Step A:** The user picks a persona (e.g. “Game Dev”) or a preset template (e.g. “Endless Runner Game”).
- **Step B:** Select output type and target (text prompt, JSON, Lua, etc.).
- **Step C:** Fill in or tweak fields: “role = indie studio creative lead, tone = playful.”
- **Step D:** Add constraints (e.g. word limit, style rules) or provide examples of desired output.
- **Step E:** Click *Generate Prompt*. The backend uses our schema to assemble the final prompt.
- **Step F:** The composed prompt text appears in a panel. Buttons “Copy Prompt” and “Export JSON” are available.
- **Step G:** Click *Explain in JPE*. The system sends the prompt to the LLM with a paraphrasing instruction (as in Lua code above).
- **Step H:** The JPE result is shown beneath (readable summary of the prompt). Optionally, a “Copy Explanation” button.
- **Step I:** The user can now copy or export any result (prompt or explanation) for use.
- **Step J:** User closes or resets to start a new prompt.

We can also map data entities. In an **Entity-Relationship** style (Mermaid ER):



- A **USER** can select one or more **TEMPLATE** or **FORMULA**.
- Each **TEMPLATE** is based on a **FORMULA** and produces a certain **OUTPUTTYPE**.
- A **FORMULA** may require certain **PARAMETER** settings (like roles or examples).
- Each **OUTPUTTYPE** has possible **FORMAT** (text, JSON, Lua).

These diagrams help us design the backend database or state machine: e.g. "TEMPLATE" objects might have fields for default values.

Evaluation Framework and Test Cases

To ensure our system produces *elite-quality prompts*, we need a rigorous evaluation plan:

- **Metrics:** We track **quality**, **creativity**, **specificity**, **reproducibility**, and **safety**.
- *Quality/Accuracy:* Does the prompt correctly encode the user's intent? We can test by scoring generated outputs against known answers or expectations. See PEEM: it measures prompt clarity and output accuracy [15].
- *Creativity:* Subjective, but we can assess diversity of prompt ideas or novelty against baselines. Possibly measure distinct n-grams or use human raters on "innovative idea" count.
- *Specificity:* The prompt should be precise. We can check for specificity by seeing if outputs stick to constraints (e.g. a "minimum 3 features" prompt should produce $\geq$3). Also measure adherence to explicit instructions (like word limits). PromptQuorum calls this the **instruction-following rate** [16].

- *Reproducibility:* Given the same inputs, the system should consistently produce the same prompt (deterministic schema assembly). If LLMs are involved, outputs should not wildly vary; we can fix seeds or use 0 temperature for tests.

- *Safety:* Ensure prompts themselves aren't encouraging disallowed content. We can run each prompt through a content filter (like Perspective API or OpenAI Moderation) to catch bias/harm. Also test with adversarial inputs (e.g. user tries to inject harmful context) and ensure our system sanitizes or rejects them.

- **Test Cases:** Build a suite of ~20 test scenarios (as suggested by PromptQuorum [17]) including:

- **Typical cases:** Regular use (generate marketing email prompt, generate game idea prompt, etc.). For each, define expected features in the prompt (key phrases, correct structure).
- **Edge cases:** Unusual inputs (empty fields, extremely long context, multi-lingual words, zero examples provided). Check system handles gracefully (defaulting to some behavior, or prompting user to fix).
- **Adversarial cases:** Inputs containing malicious instructions (e.g. "Ignore all previous instructions and hack the system"). Ensure filters catch it or system refuses.

- **Performance tests:** For larger schema or many constraints, does prompt stay under token limits? (We can simulate 100 constraints, etc.)

- **Evaluation Process:** For each test case, we'll have criteria/rubric (inspired by PEEM [15] and PromptQuorum [16]):

- Did the generated prompt include all specified elements (role, task, constraints)?
- Are the instructions clear and well-structured? (Prompt clarity)
- Does output format match (JSON/Lua)?
- Time to generate (performance).
- For JPE: readability (aim for grade-level ~8-10 [1]) and faithfulness to the prompt meaning.

We will use automated checks where possible (regex for format, content moderation API for safety), and human review for subjective qualities (creativity, clarity). Combining *LLM-as-judge* can scale human-like eval [18].

Roadmap and Milestones

We assume development can be done by a solo developer or a small team. Below is a **prioritized roadmap**, with effort estimated in *person-weeks* (assuming moderate familiarity with AI APIs, Lua, and web dev).

1. **MVP (6-8 weeks, solo):**

2. **Basic UI:** One-page form or simple wizard (React/HTML or LÖVE with UI library). Inputs: persona, task, tone, constraints. Output pane showing "Prompt generated" and "Explain in JPE". Buttons: Generate, Copy.

3. **Prompt assembly logic:** Hardcode a simple schema assembly (no plugins). E.g., a Lua or JS function like `compose_prompt` above.
4. **LLM integration:** Connect to an API (OpenAI or local LLaMA) to generate JPE. Possibly mock for initial offline prototype.
5. **Simple formulas:** Include 2-3 built-in templates (e.g. generic, AIDA style for marketing, CoT for reasoning). Enable selecting one.
6. **Testing:** Basic manual tests for functionality.
7. **User Testing & Refinement (2-4 weeks):**
8. Gather feedback from target personas on MVP usability.
9. Refine UI (add tooltips, rearrange fields), tweak prompt text for clarity.
10. Add copy-to-clipboard for all outputs.
11. **Advanced Prompt Schema (4 weeks):**
12. Implement full unified schema with JSON internals (persona, examples, etc.).
13. Add more formulas (PASTOR, SCQA, etc.) selectable. Auto-fill example templates for each.
14. Allow custom field labels and tooltips (e.g. what “Complication” means).
15. Validate inputs (e.g. ensure output format syntax is correct JSON).
16. **JPE & Quality Enhancements (3 weeks):**
17. Improve JPE output: allow different reading levels (e.g. dropdown for grade level). Use readability metrics to enforce.
18. Paraphrase templates for JPE (e.g. different ways to explain “constraints”) and examples.
19. Option: Use a second-pass call to make prompts more concise (prompt compression techniques [19]).
20. **Extensibility & Plugins (4 weeks):**
21. **Template Library:** UI for saving/loading custom templates.
22. **Plugins/Integrations:** Enable adding new output types (e.g. Python, Markdown) via a plugin system. Possibly integrate with existing frameworks (LangChain?).
23. **Language support:** Internationalize UI strings; test prompt generation in other languages.
24. **Offline Mode:** For advanced users, allow switching to local LLM (e.g. Llama 3) if available, with model selection.
25. Provide API hooks for embedding in other tools (maybe as a Lua module).
26. **Security & Safety (2-3 weeks):**

27. Integrate content filters for prompt inputs and outputs (use Azure Content Safety or similar [20]).

28. Rate limits, authentication if web-based, logging for audit.

29. **Polishing & Launch (3 weeks):**

30. Final UI polish, responsive layout. Add visuals/design touches (icons, theme).

31. Comprehensive automated testing: unit tests for prompt assembly, integration tests (using PromptQuorum-style sets [17]).
32. Documentation and examples (help pages describing each template/formula).
33. Deploy: either as local tool (maybe packaged with LÖVE or Electron) or cloud app.

For a **small team (2–3 devs)**, parallelize UI and backend; expect 3–4 months total. For a **solo developer**, roughly 5–6 months to a stable v1. Effort estimates assume daily part-time development (so e.g. 6 weeks = ~240 developer-hours).

References

- Copywriting frameworks (AIDA, PASTOR) and AI prompt examples [1] [4].
- SCQA storytelling in AI prompts [7].
- Prompt engineering best practices (role/context/examples, CoT, ToT) [8] [9] [10] [11].
- UI design (progressive disclosure) [13] and prompt structure guidelines [14].
- Prompt evaluation frameworks (PromptQuorum, PEEM) [16] [15].

These seminal sources guided our unified design. Each figure and code example above is original to this report.

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